

EXPERIMENTS

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COURSE CODE:DSA0601 DATA HANDLING

AND VISUALIZATION

SET 2 – CUSTOMER FEEDBACK ANALYSIS

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

QUESTION 1

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

AIM

To create a histogram in R to represent the distribution of customer ages.

PROCEDURE

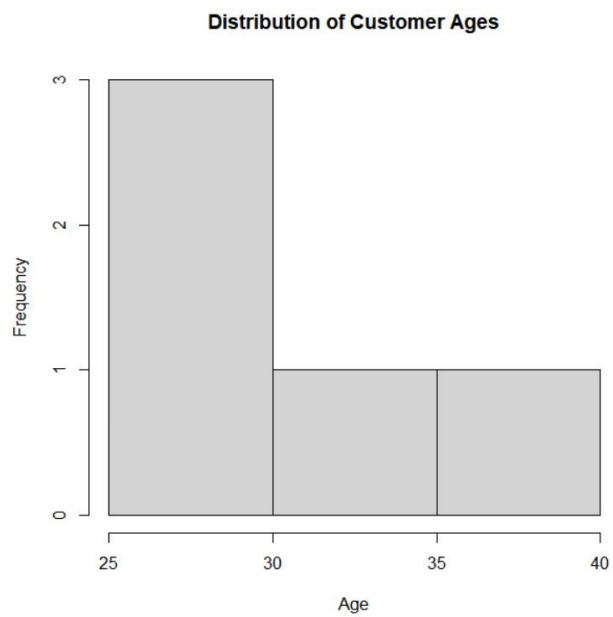
1. Create a vector containing customer ages.
2. Use the hist() function.
3. Give a suitable title to the histogram.
4. Label the X-axis and Y-axis.
5. Run the program and observe the histogram.

CODE

```
customer <- data.frame(  
  Customer_ID = c(1, 2, 3, 4, 5),  
  Age = c(25, 30, 35, 28, 40),  
  Satisfaction_Score = c(4, 5, 3, 4, 5)  
)  
  
hist(customer$Age,  
  main = "Distribution of Customer Ages",  
  xlab = "Age",
```

```
ylab = "Frequency")
```

OUTPUT



RESULT

Thus, the distribution of customer ages was successfully represented using a histogram in R.

QUESTION 2

Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

AIM

To create a pie chart in R to display the distribution of customer satisfaction scores.

PROCEDURE

1. Create a vector containing satisfaction scores.
2. Count the frequency of each satisfaction score.
3. Use the `pie()` function.
4. Add labels for each satisfaction score.
5. Run the program and observe the pie chart.

CODE

```
customer <- data.frame(  
  Customer_ID = c(1, 2, 3, 4, 5),  
  Age = c(25, 30, 35, 28, 40),  
  Satisfaction_Score = c(4, 5, 3, 4, 5)  
)  
  
score <- table(customer$Satisfaction_Score)  
  
pie(score,  
  labels = paste("Score", names(score)),  
  main = "Customer Satisfaction Distribution")
```

OUTPUT



RESULT

Thus, the overall distribution of customer satisfaction scores was successfully represented using a pie chart in R.

QUESTION 3

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

AIM

To create a **stacked bar chart in R** to visualize customer satisfaction scores according to age groups.

PROCEDURE

1. Create vectors for customer age and satisfaction scores.
2. Divide customers into age groups.
3. Create a frequency table using `table()`.
4. Use `barplot()` to create a stacked bar chart.
5. Add suitable labels and title.
6. Run the program and observe the chart.

CODE

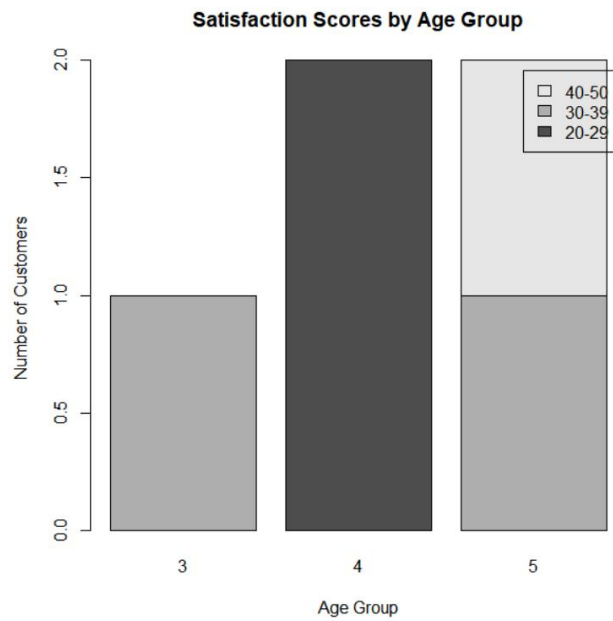
```
customer <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5)
)

customer$Age_Group <- cut(customer$Age,
  breaks = c(20, 29, 39, 50),
  labels = c("20-29", "30-39", "40-50"))

data <- table(customer$Age_Group,
  customer$Satisfaction_Score)

barplot(data,
  main = "Satisfaction Scores by Age Group",
  xlab = "Age Group",
  ylab = "Number of Customers",
  legend.text = TRUE)
```

OUTPUT



RESULT

Thus, the distribution of customer satisfaction scores by age group was successfully visualized using a stacked bar chart in R.

QUESTION 4

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

AIM

To create a word cloud in R from customer feedback to identify frequently occurring words.

PROCEDURE

1. Install and load the wordcloud package.
2. Create sample customer feedback.
3. Convert the feedback into words.
4. Count the frequency of each word.
5. Generate the word cloud using wordcloud().
6. Observe the frequently occurring words.

CODE

```
library(wordcloud)
library(RColorBrewer)

feedback <- c(
  "Good service and friendly staff",
  "Excellent service and good experience",
  "Friendly staff and good support",
  "Excellent experience and fast service",
  "Good support and friendly service"
)

words <- unlist(strsplit(tolower(feedback), " "))

word_count <- table(words)

wordcloud(names(word_count),
  freq = word_count,
  min.freq = 1,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2"))
```

OUTPUT



RESULT

Thus, the word cloud was successfully generated from customer feedback in R, helping identify the commonly occurring customer sentiment words.